

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: (II)

Problem Set 4

(MOME, MMSE Estimator, MLE)

1. Let $X_1, X_2, \dots, X_n$ be a random sample from the following distributions with given p.m.f./ p.d.f. Find the method of moments estimator (MOME) of each $g(\theta)$:

a) $f_\theta(x) = \begin{cases} \frac{e^{-\theta}\theta^x}{x!}, & \text{if } x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$ for $g(\theta) = e^\theta$ and $g(\theta) = P_\theta(X_1 + X_2 + X_3 = 1)$, $\theta = (0, \infty)$

b) $f_\theta(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} e^{-x/\beta} x^{\alpha-1}, & \text{if } x > 0 \\ 0, & \text{otherwise,} \end{cases}$ for $g(\theta) = (\alpha, \beta)'$, $\theta = (\alpha, \beta)'$ $\theta = (0, \infty) \times (0, \infty)$

c) $f_\theta(x) = \begin{cases} \frac{e^{-x/\theta}}{\theta}, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases}$ for $g(\theta) = P_\theta(X_1 \leq 1)$, $\theta = (0, \infty)$

d) $X \sim U(-\theta/2, \theta/2)$, for $g(\theta) = (1 + \theta)^{-1}$, $\theta = (0, \infty)$.

2. Let $X_1, X_2, \dots, X_n$ be a random sample from $U(0, \theta)$. Define class of estimators $\delta_c(\mathbf{X}) = \{cX_{(n)}; X_{(n)} = \max X_i, 1 \leq i \leq n\}$. Obtain the MMSE estimator of θ in this class.

3. Let X be a discrete random variable with the following p.m.f. $0 \leq \theta \leq 1$ is an unknown parameter.

$X = x$	0	1	2	3
$P(X = x)$	$2\theta/3$	$\theta/3$	$2(1 - \theta)/3$	$(1 - \theta)/3$

Ten observations, viz., 3, 0, 2, 1, 3, 2, 1, 0, 2, 1 were sampled from the distribution. What is the maximum likelihood estimate of θ ?

4. Let $X_1, X_2, \dots, X_n$ be a random sample from each of the following distributions with the given p.m.f./p.d.f. Find the maximum likelihood estimator (MLE) of θ in each case:

a) $f_\theta(x) = \begin{cases} \frac{e^{-\theta}\theta^x}{x!}, & \text{if } x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$

b) $f_\theta(x) = \begin{cases} \theta x^{\theta-1}, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$

c) $f_\theta(x) = \begin{cases} \frac{e^{-|x-\theta|}}{2}, & \text{if } -\infty < x < \infty \\ 0, & \text{otherwise.} \end{cases}$

d) $X_i \sim U(-\theta/2, \theta/2)$, $i = 1(1)n$

5. Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with the following p.d.f. Find the MLEs of θ_1 and θ_2 :

a)
$$f_{\theta_1, \theta_2}(x) = \begin{cases} \frac{e^{-(x-\theta_1)/\theta_2}}{\theta_2}, & \text{if } x \geq \theta_1 \\ 0, & \text{otherwise.} \end{cases}$$

b)
$$f_{\theta_1, \theta_2}(x) = \begin{cases} \frac{\theta_2^{\theta_1}}{\Gamma(\theta_2)} e^{-\theta_2 x} x^{\theta_1-1}, & \text{if } x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

c)
$$f_{\theta_1, \theta_2}(x) = \begin{cases} \frac{1}{2\sqrt{3}\theta_2}, & \text{if } \theta_1 - \sqrt{3}\theta_2 < x < \theta_1 + \sqrt{3}\theta_2 \\ 0, & \text{otherwise} \end{cases}$$

6. Let $X_1, X_2, \dots, X_n$ be a random sample from a discrete distribution with the following p.m.f. Find the MLE of N .

a)
$$f_N(x) = \begin{cases} \frac{1}{N}, & \text{if } x = 1(1)N \\ 0, & \text{otherwise.} \end{cases}$$

b)
$$f_N(x) = \begin{cases} \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}, & \text{if } \max(0, n - N + M) \leq x \leq \min(n, M) \\ 0, & \text{otherwise.} \end{cases}$$

7. A salesman of used cars is willing to assume that the number of sales he makes per day is a Poisson random variable with parameter μ . Over the past 30 days he made no sales on 20 days and one or more sales on each of the remaining 10 days. Find the maximum likelihood estimate of μ .
8. Let $X_1, X_2, \dots, X_n$ be a random sample from $N(\theta, 1)$, $\theta \geq 0$. Find the MLE of θ .
9. Let X be a random observation from $\text{Bin}\left(1, \frac{1}{1+e^\theta}\right)$, $\theta \in \mathfrak{R}$. Find the MLE of θ .
10. Let $X_1, X_2, \dots, X_n$ be a random sample from $P(\theta)$. Find a consistent estimator of $\theta^3(3\sqrt{\theta} + \theta + 12)$.
11. Let $X_1, X_2, \dots, X_n$ be a random sample from $U(0, \theta)$. Find a consistent estimator of e^θ .
12. Let $X_1, X_2, \dots, X_n$ be a random sample from $\text{Poisson}(\theta)$. Find the asymptotic distribution of the MLE of $e^{-\theta}$.
13. Let $X_1, X_2, \dots, X_n$ be a random sample from $N\left(0, \frac{1}{\theta}\right)$. Find the asymptotic distribution of the MLE of $\frac{1}{\theta}$.